Inflation Persistence After an Energy-Price Shock: A Two-Sector New Keynesian Model with Production Networks and Real Wage Rigidity

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EC424 Extended Essay

Observations in the UK

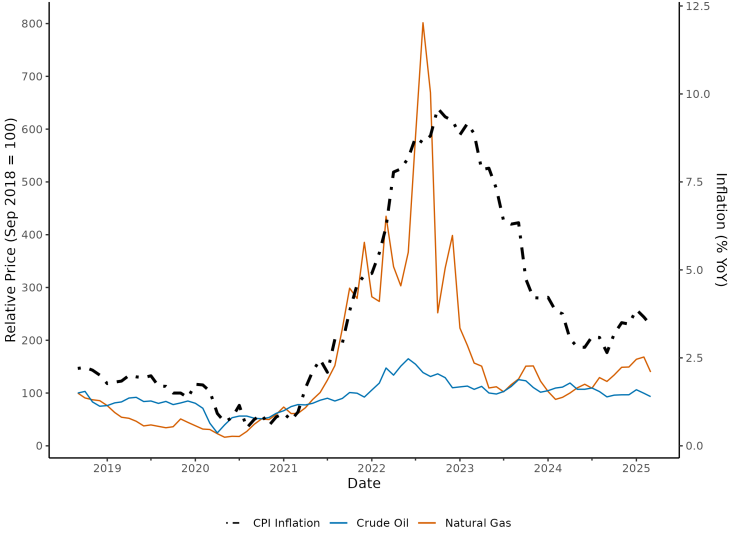

Figure 1: Energy Prices and UK CPI Inflation (Sep 2018 - Mar 2025)

What happened in the UK after the gas price spike?

- Natural gas price decreased relatively quickly; CPI inflation decreased **more slowly**.
- **Few months lag** between the peak of gas price and the peak of CPI inflation.
- CPI inflation did not return to the pre-shock level[1]

[1] I do not feature this in my model in fear of simultaneous trends.

Three Model Extensions from the Baseline

1. **Production network**
   - Structured vertically with upstream and downstream industries
     - Upstream industry: energy + labour $\longrightarrow$ intermediate goods
     - Downstream industry: intermediate goods + labour $\longrightarrow$ final consumption goods
   - Production function remains Leontief for both sectors
   - Competitive sector remains unchanged, producing tradable goods using labour only for the balance of payments equilibrium
   - Single labour market with free mobility ($\Longrightarrow$ equal wage across sectors)

2. **Myopia** in downstream firms' Calvo reset price determination

- Minton and Wheaton (2023); Gabaix (2020)
- Firms are reluctant / difficult to forecast the future, and thus discount the future expectations
- The myopic expectation of $\hat{v}_{t+k}$ (in log-linearised form) is

$$\tilde{\mathbb{E}}_t[\hat{v}_{t+k}] \equiv m_f^k \mathbb{E}_t[\hat{v}_{t+k}],$$

- where $m_f \in [0, 1]$ with $m_f = 1$ being completely rational

3. **Real wage rigidity** in the labour market

- Blanchard and Galí (2007)
- Only a fraction $1 - \psi_w$ of the gap between the actual real wage $\hat{w}_t$ and the flexible real wage $\hat{w}_t^f$ is adjusted in each period:

$$\hat{w}_t = \psi_w \hat{w}_{t-1} + (1 - \psi_w)\hat{w}_t^f,$$

- where $\psi_w \in [0, 1]$ with $\psi_w = 0$ being completely flexible

Log-Linearised Model Equations 1/2

$$\hat{y}_t = \hat{L}_t - \alpha^U \alpha^D \hat{s}_t \qquad \text{(aggregate production function)}$$

$$\hat{x}_t^D = \Phi_{DU}(\hat{p}_t^U - \hat{p}_t^D) + (1 - \Phi_{DU})\hat{w}_t \qquad \text{(downstream real marginal cost)}$$

$$\hat{X}_t^D = \hat{x}_t^D + \hat{p}_t^D \qquad \text{(downstream nominal marginal cost)}$$

$$\hat{X}_t^U = \alpha^U \hat{s}_t + \hat{w}_t + \hat{p}_t^D \qquad \text{(upstream nominal marginal cost)}$$

$$\hat{w}_t^f = \frac{1}{\sigma}\hat{y}_t + \frac{1}{\psi}\hat{L}_t \qquad \text{(flexible real wage)}$$

$$\hat{w}_t = \psi_w \hat{w}_{t-1} + (1 - \psi_w)\hat{w}_t^f \qquad \text{(real wage rigidity)}$$

Log-Linearised Model Equations 2/2

$$\hat{\pi}_t = \hat{p}_t^D - \hat{p}_{t-1}^D \qquad \text{(CPI inflation)}$$

$$\hat{p}_t^D = \theta^D \hat{p}_{t-1}^D + (1-\theta^D)\hat{p}_t^{D*} \qquad \text{(downstream Calvo price level)}$$

$$\hat{p}_t^{D*} = \beta m_f \theta^D \mathbb{E}_t[\hat{p}_{t+1}^{D*}] + (1-\beta m_f \theta^D)\hat{X}_t^D \qquad \text{(downstream reset price)}$$

$$\hat{p}_t^U = \theta^U \hat{p}_{t-1}^U + (1-\theta^U)\hat{p}_t^{U*} \qquad \text{(upstream Calvo price level)}$$

$$\hat{p}_t^{U*} = \beta\theta^U \mathbb{E}_t[\hat{p}_{t+1}^{U*}] + (1-\beta\theta^U)\hat{X}_t^U \qquad \text{(upstream reset price)}$$

$$\hat{y}_t = \mathbb{E}_t[\hat{y}_{t+1}] - \sigma(\hat{i}_t - \mathbb{E}_t[\hat{\pi}_{t+1}]) \qquad \text{(IS equation)}$$

$$\hat{i}_t = \phi\hat{i}_{t-1} + (1-\phi)(\chi_\pi \hat{\pi}_t + \chi_y \hat{y}_t) \qquad \text{(Taylor rule)}$$

$$\hat{s}_t = \rho\hat{s}_{t-1} + u_t \qquad \text{(Terms of trade AR(1) process)}$$

Parameter Calibration

Parameter	Symbol	Value
Households		
Discount factor	β	0.9984
Elasticity of intertemporal substitution	σ	0.5
Frisch elasticity of labour supply	ψ	3
Production and cost structure		
Energy cost share (upstream)	α^U	0.375
Intermediate input share (downstream)	α^D	0.4
Elasticity of substitution	ε	11
Downstream's intermediate good-cost share	Φ_{DU}	0.4231 (implied)
Price setting		
Downstream Calvo probability	θ^D	11/12
Upstream Calvo probability	θ^U	3/4
Myopia parameter (downstream)	m_f	0.5
Labour market		
Real wage rigidity	ψ_w	$0.7^{1/3} \approx 0.888$
Monetary policy		
Interest-rate smoothing	ϕ	0.9
Inflation response	χ_π	3
Output response	χ_y	0.5
Shock process		
Terms of trade persistence	ρ	0.8

Energy cost share (upstream): $\alpha^U = 0.375$

- Output-weighted ave. energy cost share across eight most energy-intensive sectors[2]
  - Model's weak point: those sectors produce final consumption goods in reality
- Based on the input-output table of the UK economy 2017-2022 by ONS

Intermediate input share (downstream): $\alpha^D = 0.4$

- So that the energy cost share in the final consumption goods is $\alpha^U \alpha^D = 0.15$

Downstream's intermediate good-cost share: $\Phi_{DU} = 0.4231$ (implied)

$$\Phi_{DU} = \frac{\alpha^D \mu}{1 - \alpha^D + \alpha^D \mu} = 0.4231$$

- Cost-based share of intermediate goods in the downstream real marginal cost
- Takes into account the upstream industry's markup on the intermediate goods
  - cf. $\alpha^D = 0.4$ does not

[2] Electric Power Generation, Gas Manufacture & Distribution, Industrial Gases, Inorganics, Fertilisers, Petrochemicals, Paper and Paper Products, Glass, Ceramics, Stone, Basic Iron and Steel, and Other Basic Metals & Casting

Upstream Calvo probability: $\theta^U = 3/4$

- Average price duration of 4 months
- Nakamura and Steinsson (2008): prices for intermediate goods are more flexible than those for final consumption goods
- Sensitivity analysis with $\theta^U \in \{0.5, 3/4, 11/12\}$

Myopia parameter (downstream): $m_f = 0.5$

- The mid-point of the parameter space $[0, 1]$ given the variety of intermediate goods
- Sensitivity analysis with $m_f \in \{0, 0.5, 1\}$

Real wage rigidity: $\psi_w = 0.7^{1/3} \approx 0.888$

- The mid point of the two sample values in Blanchard and Galí (2007) in monthly frequency
- Sensitivity analysis with $\psi_w \in \{0, 0.7^{1/3}, 0.9^{1/3}\}$

Other parameters are calibrated to standard values in the literature.

Results

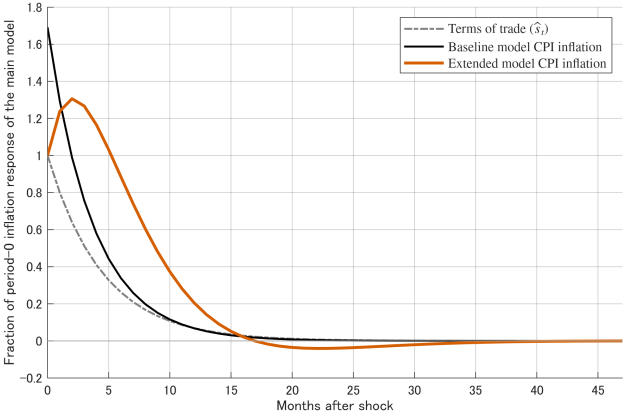

Figure 2: Inflation Persistence vs Energy Price Shock Decay

- CPI inflation of both models is normalised based on my model's initial response

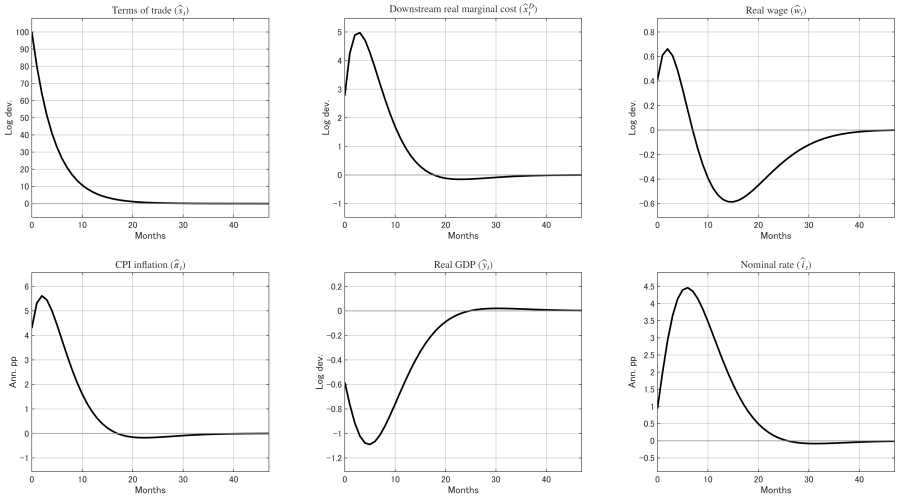

Figure 3: Impulse Responses of the Model to an Exogenous Terms of Trade Shock

Downstream Real Marginal Cost $\hat{x}_t^D$

- peaks at 5.0% with a three-month lag
- Lag is due to the upstream Calvo pricing + real wage rigidity

Real Wage $\hat{w}_t$

- High $\hat{s}_t$ demands more labour for the competitive sector
  - This offsets the negative effect due to the productivity decrease
- After the peak, nominal interest rate will be raised to stabilise inflation, which decreases labour demand
  - But adjustment is slow due to real wage rigidity

CPI Inflation $\hat{\pi}_t$

- Peaks at 5.6% with a three-month lag
  - Myopia of the downstream firms contributes to the lag and persistence

Sensitivity Analysis (IRFs in Appendix)

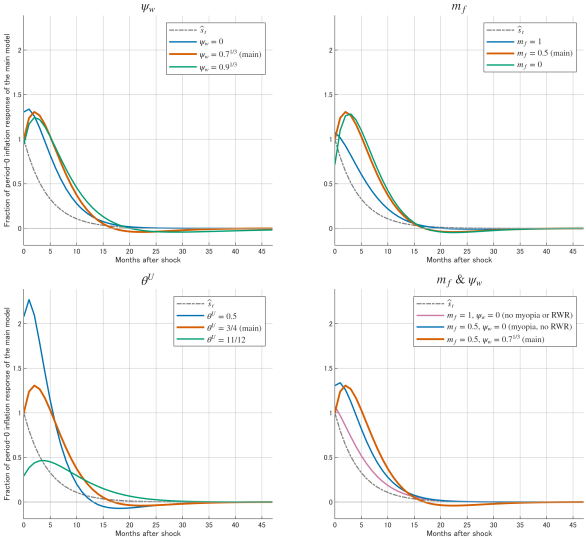

Figure 4: Sensitivity Analysis Results

Real Wage Rigidity ψ_w

- No real wage rigidity ($\psi_w = 0$):
  - Inflation responds immediately with initial response almost as high as its peak in one month
- Higher real wage rigidity ($\psi_w = 0.9^{1/3}$):
  - Slightly smaller response peak
  - But inflation is more persistent with a slower decay

Downstream Firm Myopia m_f

- No myopia / completely rational ($m_f = 1$):
  - Inflation decreases monotonically after the initial shock
  - Correct expectations that energy price will only decrease back to the steady state level
- Full myopia ($m_f = 0$):
  - Initial inflation response is smaller
  - but catches up in four months to reach the almost same peak

Upstream Calvo Probability θ^U

Combination: Myopia m_f and Real Wage Rigidity ψ_w

References

Blanchard, Olivier, and Jordi Galí. 2007. "Real Wage Rigidities and the New Keynesian Model." *Journal of Money, Credit and Banking* 39 (s1): 35–65. https://doi.org/10.1111/j.1538-4616.2007.00015.x.

Gabaix, Xavier. 2020. "A Behavioral New Keynesian Model." *American Economic Review* 110 (8): 2271–2327. https://doi.org/10.1257/aer.20162005.

Minton, Robert, and Brian Wheaton. 2023. "Delayed Inflation in Supply Chains: Theory and Evidence." *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.4470302.

Nakamura, Emi, and Jón Steinsson. 2008. "Five Facts about Prices: A Reevaluation of Menu Cost Models." *Quarterly Journal of Economics* 123 (4): 1415–64. https://doi.org/10.1162/qjec.2008.123.4.1415.

Appendix

The Baseline Model

$$\hat{y}_t = \hat{L}_t - \alpha \hat{s}_t \qquad \text{(aggregate production function)}$$

$$\hat{x}_t = \hat{w}_t + \alpha \hat{s}_t \qquad \text{(real marginal cost)}$$

$$\hat{X}_t = \hat{x}_t + \hat{P}_t \qquad \text{(nominal marginal cost)}$$

$$\hat{w}_t = \frac{1}{\sigma}\hat{y}_t + \frac{1}{\psi}\hat{L}_t \qquad \text{(labour supply)}$$

$$\hat{\pi}_t = \hat{P}_t - \hat{P}_{t-1} \qquad \text{(inflation)}$$

$$\hat{P}_t = \theta \hat{P}_{t-1} + (1-\theta)\hat{P}_t^* \qquad \text{(Calvo price level)}$$

$$\hat{P}_t^* = \beta\theta \mathbb{E}_t[\hat{P}_{t+1}^*] + (1-\beta\theta)\hat{X}_t \qquad \text{(reset price)}$$

$$\hat{y}_t = \mathbb{E}_t[\hat{y}_{t+1}] - \sigma(\hat{i}_t - \mathbb{E}_t[\hat{\pi}_{t+1}]) \qquad \text{(IS equation)}$$

$$\hat{i}_t = \phi \hat{i}_{t-1} + (1-\phi)(\chi_\pi \hat{\pi}_t + \chi_y \hat{y}_t) \qquad \text{(Taylor rule)}$$

$$\hat{s}_t = \rho \hat{s}_{t-1} + u_t \qquad \text{(terms-of-trade shock)}$$

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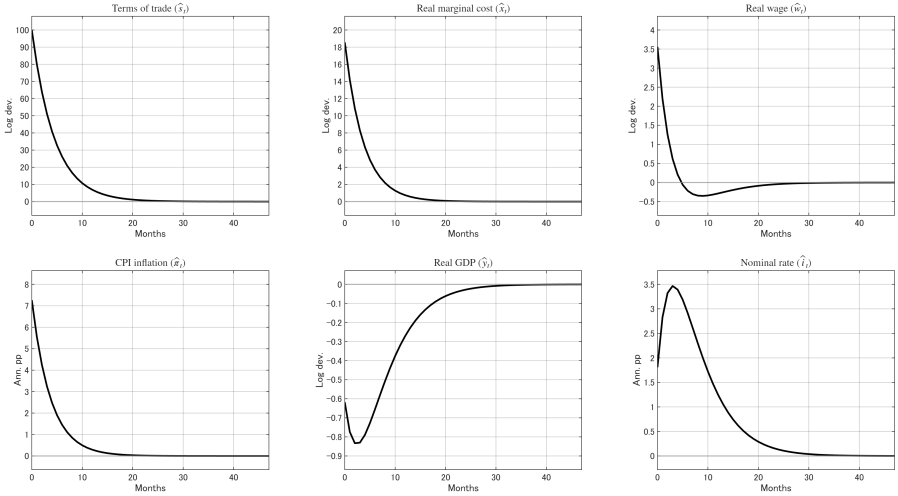

Figure 5: Impulse Responses of the Baseline Model to an Exogenous Terms of Trade Shock

Impulse Responses from Sensitivity Analysis

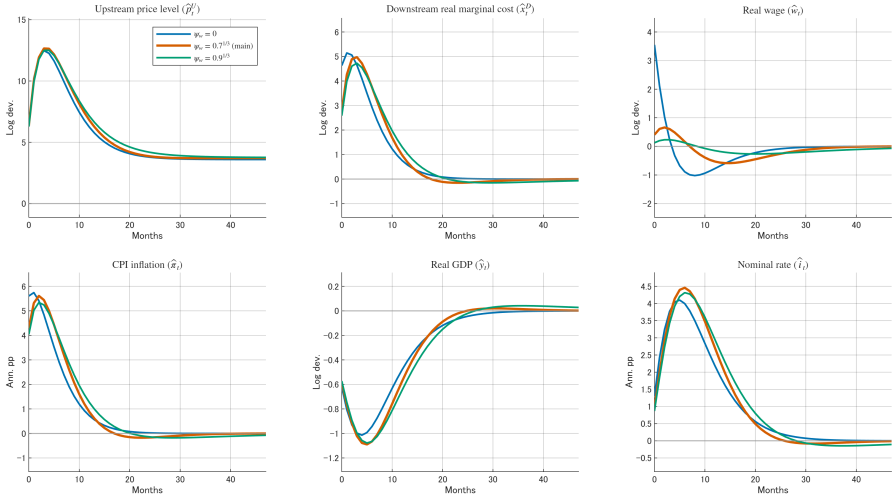

Figure 6: Impulse Responses for Varying Real Wage Rigidity ψ_w

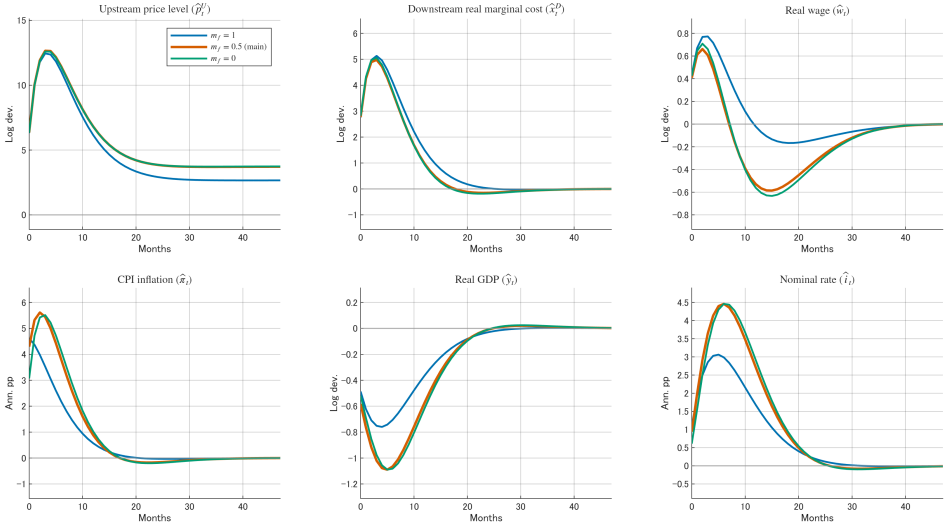

Figure 7: Impulse Responses for Varying Myopia m_f

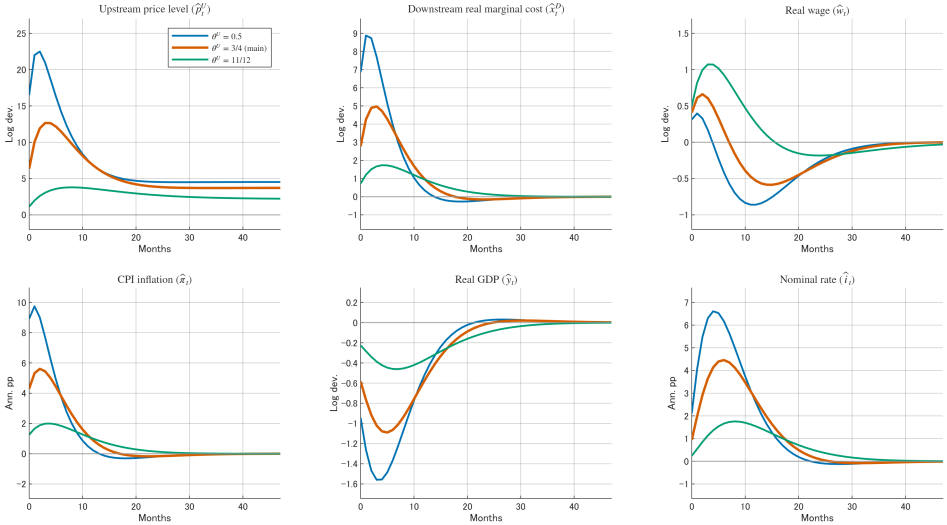

Figure 8: Impulse Responses for Varying Upstream Calvo Probability θ^U

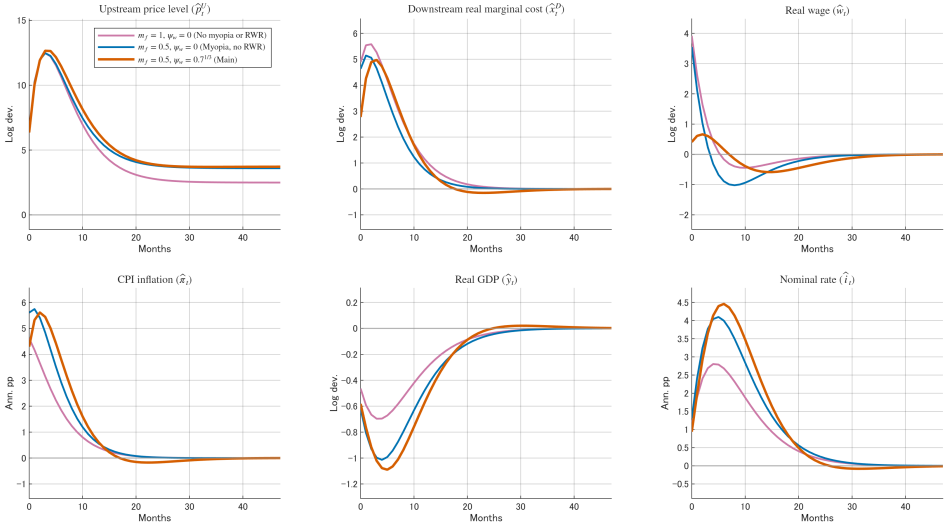

Figure 9: Impulse Responses for Different Combinations of m_f and ψ_w